

Mathematics A

Unit 1 • Chapter OL-T44 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

6 classified items • 28 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T44 • Expertise Q16+ • 6 items • 28 marks

Chapter	Items	Marks	Starts
01. OL-T44 Circles, Arcs & Sectors	6	28	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circles, Arcs & Sectors

Unit 1 • Expertise • 28 marks • 6 item(s)

November 2023 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

June 2023 | 4MA1-1H Q22 | 6 marks

Solve an algebraic circle-measure constraint

November 2024 | 4MA1-1H Q16 | 4 marks

Find sector area

November 2024 | 4MA1-2H Q25 | 5 marks

Find annular, overlapping or multi-circle area

January 2022 | 4MA1-1H Q16 | 4 marks

Recover radius or central angle from an arc/sector measure

January 2023 | 1H Q21 | 5 marks

Find a circular segment area or perimeter

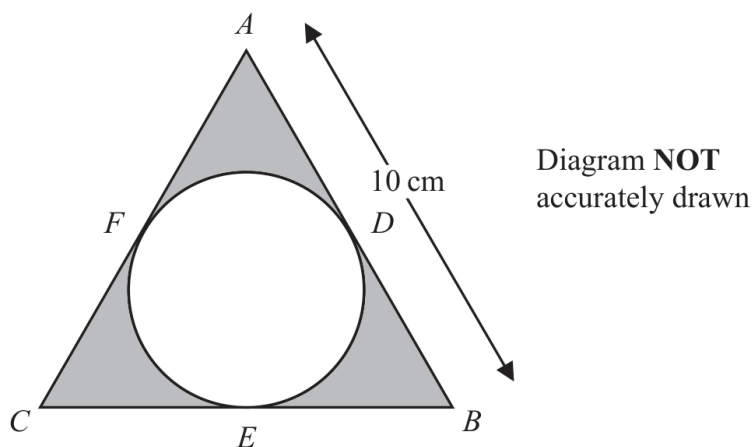
Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q20 whole

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.



D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Continue your method

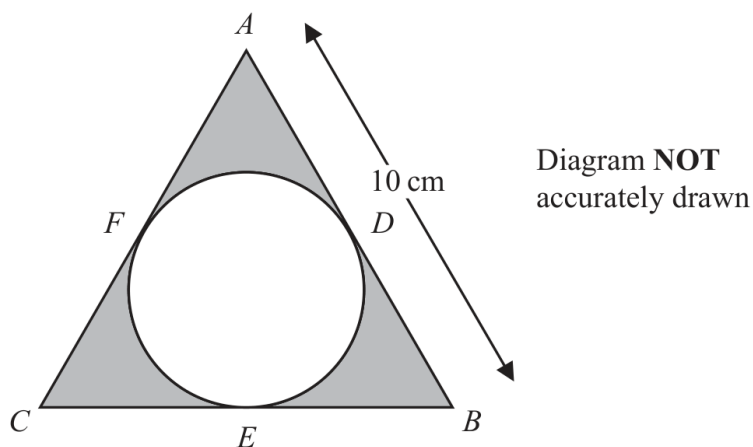
4 marks

WORKING SPACE

4MA1-2H Q20 whole

4 marks

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Give your answer correct to 3 significant figures.

..... cm^2

Shaded area between an equilateral triangle and its incircle

4 marks

whole – Shaded area between an equilateral triangle and its incircle

Find the triangle area

$$A_{\triangle} = \frac{1}{2}(10)(10) \sin 60^{\circ} = 25\sqrt{3}.$$

Find the inradius

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}.$$

Subtract the circle area

$$A_{\text{shaded}} = 25\sqrt{3} - \pi r^2 = 25\sqrt{3} - \frac{25\pi}{3}.$$

Round

$$A_{\text{shaded}} = 17.121 \dots \approx 17.1 \text{ cm}^2.$$

Final answer17.1 cm²

4MA1-1H Q22

6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

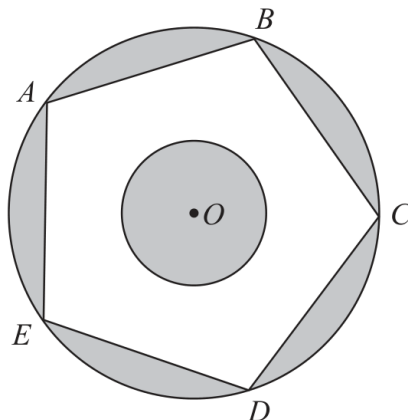


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Continue your method**6 marks**

WORKING SPACE

4MA1-1H Q22

6 marks

- 22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

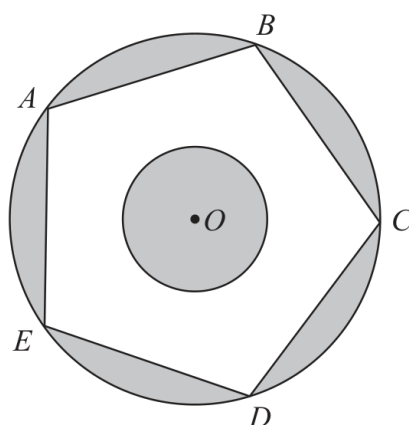


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The pentagon has sides of length 8 cm.

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$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Equal shaded and unshaded areas

6 marks

Find the circumradius

$$R = \frac{4}{\sin 36^\circ} = 6.805 \dots$$

Find the two main areas

$$A_{\text{circle}} = \pi R^2 = 145.489 \dots, \quad A_{\text{pentagon}} = 5 \left(\frac{1}{2} R^2 \sin 72^\circ \right) = 110.111 \dots$$

Use the equal-area condition

$$A_{\text{circle}} - A_{\text{pentagon}} + \pi r^2 = A_{\text{pentagon}} - \pi r^2.$$

Solve

$$r = \sqrt{\frac{2A_{\text{pentagon}} - A_{\text{circle}}}{2\pi}} = 3.44 \dots$$

Final answer

3.45 cm

4MA1-1H Q16 M01

4 marks

16 $OAPB$ is a sector of a circle, centre O

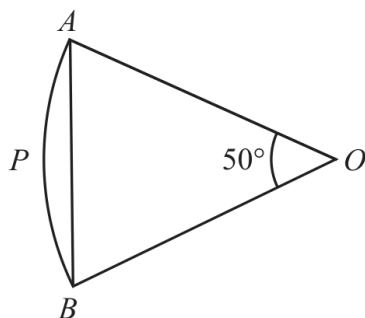


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Find the area of a sector

4 marks

M01 – Find the area of a sector

Use the triangle area

$$120 = \frac{1}{2}r^2 \sin 50^\circ \Rightarrow r^2 = 240 / \sin 50^\circ.$$

Find r

$$r = 17.7002 \dots$$

Use the sector fraction

$$A = \frac{50}{360} \pi r^2 = 136.7 \dots$$

Round

$$A = 137 \text{ cm}^2 \text{ (3 s.f.)}.$$

Final answer

$$137 \text{ cm}^2$$

4MA1-2H Q25 M01

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

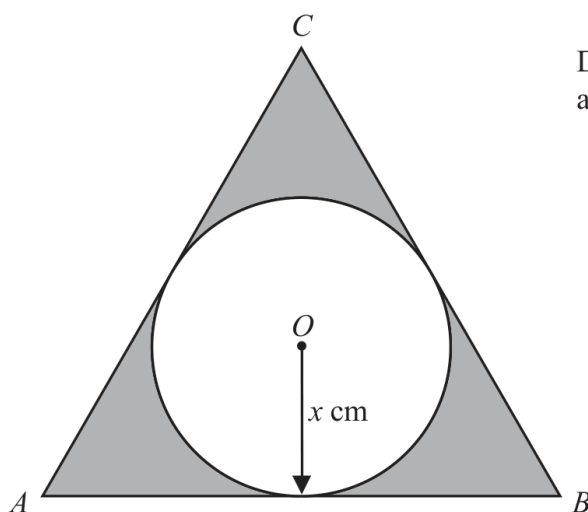


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

4MA1-2H Q25 M01

5 marks

 $n = \dots\dots\dots$

WORKING SPACE

Area of shaded regions in an equilateral triangle

5 marks

M01 – Area of shaded regions in an equilateral triangle

Find the side length

The inradius is x , so the side is $2\sqrt{3}x$.

Find triangle area

$$A_{\text{triangle}} = \frac{1}{2}(2\sqrt{3}x)(3x) = 3\sqrt{3}x^2.$$

Subtract the circle

$$A_{\text{shaded}} = (3\sqrt{3} - \pi)x^2.$$

Read n

$$n = 3\sqrt{3} - \pi = 2.0548\dots$$

Round

$$n = 2.05 \text{ (3 s.f.)}.$$

Final answer

2.05

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

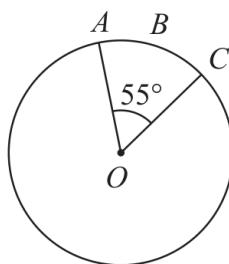


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Continue your method**4 marks**

WORKING SPACE

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

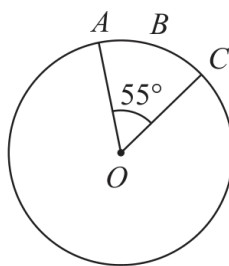


Diagram **NOT**
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A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Chapter: Circles, Arcs & Sectors

4 marks

Q16 – Chapter: Circles, Arcs & Sectors

Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure

Method

$$5 = \left(\frac{55}{360}\right) \times 2\pi r, \text{ so } r = 5 \times 360 / (55 \times 2\pi).$$

*Why: Use the arc-length formula $s = (\theta/360)2\pi r$.***Method**

$$r \approx 5.207 \text{ cm.}$$

*Why: Evaluate the radius.***Method**

$$\text{Area} = \pi r^2 \approx 85.2 \text{ cm}^2.$$

*Why: Use πr^2 .***Method**To 1 decimal place, the area is 85.2 cm^2 .*Why: Round as requested.***Final answer****area:** 85.2 cm^2

- 21 The diagram shows the cross section of a circular water pipe.

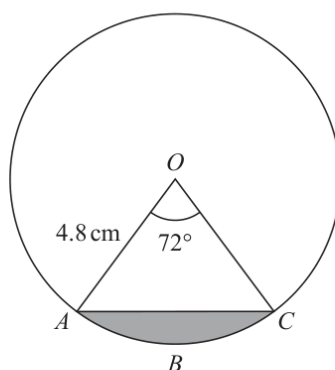


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



..... cm³

(Total for Question 21 is 5 marks)

Answer reference | January 2023 | Question 21

Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

21	$\pi \times 4.8^2 \times \frac{72}{360}$ (= 14.4(76...)) oe		5	M1 for finding the area of the sector
	$\frac{1}{2} \times 4.8^2 \times \sin 72$ (= 10.9(56...) or 11) oe or $\frac{1}{2} \times 5.6(4...) \times 3.8(8...) \text{ oe}$			M1 for finding the area of the triangle (Allow use of cosine rule/sine rule/SOHCAHTOA/Pythagoras to find AC (5.6(427.8...)) and OM (3.8(8328...)) where M is the midpoint of AC)
	"14.4(76...)" – "10.9(56...)" (= 3.520...)			M1 for finding the shaded area with all figures from correct working
	"3.5(20...)" $\times 14 \times 3 \times 60$ "3.5(20...)" $\times 2520$			M1
	Award marks within the range from correct working	8870		A1 accept 8820 – 8950 from correct working
				Total 5 marks

Worked solution

January 2023 | Question 21 | 5 marks

Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

Official final mark-scheme pages are embedded immediately after this question.